

SUPHAN BURI, Thailand

WMO: 484250

Lat: 14.474N

Lon: 100.139E

Elev: 9

StdP: 101.22

Time Zone: 7.00 (E07)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	17.8	19.3	11.7	8.6	22.1	13.3	9.6	23.1	5.4	25.2	4.8	25.3	0.7	350	0.340

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4	9.7	37.7	26.5	36.8	26.6	36.0	26.5	28.5	34.1	28.1	33.4	27.7	32.9	1.6	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
27.3	23.1	30.6	26.9	22.6	30.3	26.6	22.1	30.1	92.3	34.3	90.4	33.4	88.7	33.0	30.0

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
4.5	4.1	3.5	15.8	39.1	2.0	0.9	14.3	39.8	13.2	40.3	12.0	40.8	10.6	41.4
			13.3	29.1	1.9	0.6	12.0	29.6	10.9	29.9	9.8	30.2	8.5	30.6
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	29.0	26.4	28.3	30.0	31.4	31.0	30.2	29.6	29.5	29.2	28.8	27.7	26.1
	DBStd	2.29	2.17	1.92	1.84	1.59	1.61	1.09	1.17	1.03	1.11	1.15	1.75	2.38
	HDD10.0	0	0	0	0	0	0	0	0	0	0	0	0	0
	HDD18.3	0	0	0	0	0	0	0	0	0	0	0	0	0
	CDD10.0	6940	507	512	621	643	650	607	606	604	577	584	531	498
	CDD18.3	3898	249	279	362	393	392	357	348	346	327	326	281	240
	CDH23.3	46090	2403	3080	4507	5312	5154	4471	4139	4052	3779	3827	3048	2317
	CDH26.7	21317	943	1405	2306	3006	2741	2133	1806	1732	1579	1579	1211	875
Wind	WSAvg	0.8	0.6	0.7	0.9	0.9	0.9	1.0	1.0	0.9	0.6	0.6	0.9	0.9
Precipitation	PrecAvg	1238	8	12	30	83	185	113	109	147	309	199	33	12
	PrecMax	1576	86	43	141	227	335	199	198	237	488	381	86	72
	PrecMin	815	0	0	1	6	63	40	53	37	151	52	0	0
	PrecStd	210	22	17	37	65	97	49	48	61	112	86	30	22
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	34.1	35.7	37.6	39.2	38.9	36.7	36.2	35.8	35.3	34.6	34.4	34.1
		MCWB	24.6	25.0	25.0	26.2	27.1	27.4	26.9	26.8	27.1	27.0	26.0	25.4
	2%	DB	33.1	34.8	36.6	38.0	37.4	35.6	34.9	34.7	34.4	33.6	33.5	33.0
		MCWB	24.6	25.3	25.7	26.4	27.3	27.0	26.5	26.6	26.9	26.7	26.0	25.0
	5%	DB	32.3	33.9	35.7	37.2	36.3	34.6	33.9	33.7	33.6	32.9	32.7	32.1
		MCWB	24.2	25.2	25.8	26.5	27.3	26.9	26.5	26.4	26.7	26.5	25.6	24.5
	10%	DB	31.3	33.1	34.9	36.2	35.2	33.7	33.0	32.9	32.7	32.2	31.8	31.0
		MCWB	23.8	25.0	25.8	26.4	27.1	26.8	26.3	26.1	26.5	26.3	25.2	23.8
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	26.6	27.7	28.0	28.6	29.2	28.8	28.1	28.2	28.7	28.2	28.1	27.0
		MCDB	31.7	33.2	34.4	34.7	35.3	34.4	33.4	33.4	33.2	32.2	32.1	31.7
	2%	WB	26.0	27.0	27.4	28.1	28.5	28.2	27.5	27.4	27.8	27.7	27.3	26.2
		MCDB	31.0	32.5	33.4	33.6	34.4	33.6	32.6	32.5	32.4	31.7	31.3	30.7
	5%	WB	25.5	26.4	27.0	27.7	28.1	27.6	27.1	26.9	27.3	27.3	26.8	25.6
		MCDB	30.0	31.6	32.8	33.1	33.6	32.7	32.1	32.1	31.8	31.2	30.7	30.0
	10%	WB	25.0	25.9	26.6	27.3	27.6	27.1	26.6	26.5	26.9	26.9	26.2	25.0
		MCDB	29.0	30.7	31.9	32.7	32.9	31.9	31.4	31.3	31.2	30.8	30.0	29.4
Mean Daily Temperature Range	5% DB	MDBR	9.8	9.9	9.7	9.7	8.4	7.5	7.2	7.3	7.1	6.6	7.5	9.0
		MCDBR	10.3	10.3	10.7	10.6	9.7	8.7	8.5	8.4	8.3	7.5	8.3	9.5
		MCWBR	3.3	2.7	2.5	2.3	2.5	2.6	2.7	2.5	2.7	2.4	2.6	3.1
	5% WB	MCDBR	8.8	9.3	9.9	9.8	8.9	8.1	7.8	7.8	7.6	7.0	7.5	8.6
		MCWBR	3.0	2.7	2.6	2.6	2.8	2.7	2.8	2.7	2.8	2.5	2.6	2.9
Clear-Sky Solar Irradiance	taub	0.535	0.611	0.659	0.586	0.499	0.452	0.444	0.445	0.476	0.576	0.518	0.512	
	taud	1.950	1.800	1.721	1.928	2.231	2.359	2.384	2.368	2.273	1.948	2.079	2.033	
	Ebn at Noon	760	722	701	753	809	839	848	856	835	741	771	769	
	Edh at Noon	181	219	242	197	143	124	122	125	138	186	158	163	
All-Sky Solar Radiation	RadAvg	4.84	5.28	5.54	6.02	5.58	5.24	4.91	4.87	4.85	4.77	4.85	4.74	
	RadStd	0.28	0.21	0.40	0.32	0.42	0.33	0.29	0.34	0.29	0.28	0.29	0.20	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
	N/A	N/A	N/A	N/A	+0.60	+0.64	N/A	N/A	N/A	N/A
Regional (26 neighbors)	N/A	N/A	N/A	N/A	+0.27	+0.28	N/A	N/A	+95	+94

Nomenclature: See separate page